

An Automatic Color Correction Method Inspired By The Retinex And Opponent Colors Theories

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Abstract—Color correction indicates the process of changing the colors of a digital image in order to reach given objectives. Usually the criteria under which color correction is performed match those of white-balancing and color constancy. In this work we present an automatic color correction method inspired by the Retinex and opponent colors theories with theoretical applicability on images with multiple illuminants. The method here presented has been tested in the field of skin tone color correction.

I. INTRODUCTION

Color correction indicates the process of changing the colors of a digital image in order to reach given objectives. Most commonly color correction is used in digital imaging to imitate the human capability known as *color constancy* (also known as *chromatic adaptation*). Another common field of application is that of *image enhancement*.

Color constancy, named white balance in photography, is the ability of human beings to recognize an object color disregarding (color of) the light used to illuminate the object [1]. Such ability is not perfect, due to the limitations of the human sensorial and neural systems (see e.g. the phenomenon of metamerism), yet it is very powerful and allows the human being to adapt to a plethora of different viewing conditions. Human color constancy exhibits time and space variant behaviour that make a computational version very hard to realize.

Opposed to *human color constancy*, there is *machine color constancy* (or computational color constancy): i.e. color correction performed by some kind of computational device. The idea inspiring machine color constancy is that a computer, given enough data and an adequate model, will always be able to remove the illuminant color cast, mimicking the human visual system if desired.

More than ten years have passed since Funt et al. proved most of the machine color constancy methods available at the time to be “not good enough” [2] and even if a lot of progress has been made, color constancy still remains one of the most prominent topics of contemporary research.

This paper is organized as follows. We start with a brief excursus on past and current literature regarding the topic of color constancy (see Sec. II). In Sec. III we introduce

the problem and formulate the necessary hypotheses. Sec. IV contains the description of the proposed method and finally we explain and comment the experimental results in Sec. V.

II. PREVIOUS WORK

Many early works tried to extract information about the illuminants from simple RGB images, usually with the idea of recovering radiance and reflectance. Unfortunately this is an ill-posed and heavily under-constrained problem [3] and it is the authors’ humble opinion that, without proper radiometric information about the scene, this kind of approach should be avoided.

For an exhaustive discussion of the well established theories and methods behind computational color constancy and digital color imaging we invite the reader to refer to the excellent works by Sharma and Trussell [4], Ebner [1] and Fairchild [5]. Here we will limit ourselves to the indication of the fields and theories that have seen recent contributions to the topic.

We can start our brief discussion on the state of the art by mentioning the white-patch and grey-world assumptions, the former providing a solid grounding to the Retinex theory: many algorithms have been developed based on these assumptions even in recent years [6], [7], [8].

Gamut-mapping (gamut-constraint) methods also proliferated after a first proposal by Forsyth [9].

The field of neural networks has also seen a healthy and steady production of works, quite often more strictly related to the biological side of the human visual system than other works [10].

Other works find their roots in image statistics: an effective approach seems to be that of determining the parameters of an algorithm, or the algorithm to use, according to a pre-classification of the image [11], [12]. Statistical analysis (e.g. PCA) [13] and Bayesian inference [14] have also inspired many researchers.

Another recent approach in the field of multiple illuminants correction interprets the color correction process as a matting problem, obtaining pleasant results but still requiring human interaction [15].

In the following Section we will illustrate the problem and our assumptions about it.

III. PROBLEM FORMULATION

The general aim of a color correction method is to provide an image free of any color cast due to illumination. In other words we want every surface in the image to appear as if illuminated by white light.

Our work is focused on realizing an automatic method for correcting the influence of multiple lights by taking inspiration from, and fusing, Land's "Retinex" and Hering's "opposing colors" (opposing process) theories. To the authors' knowledge this is the first time that such a method is devised.

We decided to employ a color space that strictly adheres to Hering's theories: whereas most opponent color spaces (e.g. CIE- $L^*a^*b^*$, TIQ and YCC) present a magenta-cyan axis in place of a proper red-green one, the oRGB color space [16] properly opposes red to green and yellow to blue, and it also precisely defines the necessary transforms to and from $R'G'B'$.

Also, in order to perform color correction, we will be assuming that the following facts hold:

- 1) We can describe the image formation process through the trichromatic model

$$\mathbf{c} = f(\text{Rad} \times \text{Rfl}) \quad (1)$$

where:

- $\mathbf{c}$ is the acquired color of an object with reflectance Rfl illuminated by a light (possibly a mixture of lights) with radiance spectrum Rad
- $\mathbf{c}$ is of the form $\mathbf{x} = [x_R \ x_G \ x_B]^T$
- the function $f()$ maps the product of the illuminance and reflectance spectra into RGB values.

Note that $f()$ encompasses all the processes needed to acquire the scene and store it on an appropriate medium (e.g. a RAW file from a digital camera), thus including transformations induced by optics, medium noise, etc.

- 2) There is a suitable analysis of the RGB data that allows us to have a good estimate of luminance. It is very important to stress that we *are not trying to determine* $f()$, rather we look for all those pixels that have a high luminance and that are clustered around a representative C . Since such pixels represent metameric spectra of light sources from a numerical point of view, we assume that they should be treated in the same way, independently of the real scene illuminance or reflectance. We would also like to point out that the distinction between a true light source (i.e. an emissive body in the scene) and a reflected light spot is a matter of semantics. In the rest of this work we will use the word *light* to indicate emissive sources as well as reflected light sources.
- 3) The last assumption is that the Retinex model by Land holds (intended as in [17] and [18]), in which case we can treat each of the RGB channels separately and the maximum value on each channel represents the reference white for the whole image. Yet, color correction will be performed locally.

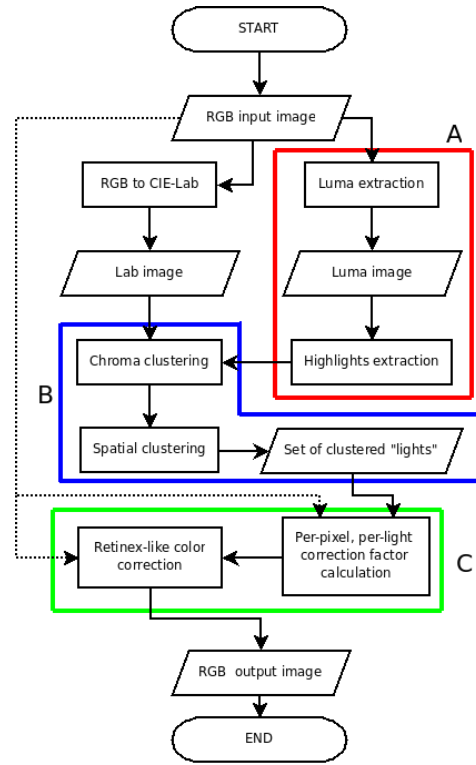


Fig. 1. Algorithm flowchart. The steps marked in color, and indicated by letters A, B and C, are treated in Sections IV-A, IV-B and IV-C respectively.

IV. PROPOSED METHOD

The flow of our algorithm is depicted in Fig. 1 and can be roughly summarized as follows:

- 1) Identification of the areas of the image that will act as lights for the color correction (see Sec. IV-A);
- 2) Chromatic and spatial clustering of the light identified in the previous step (see Sec. IV-B);
- 3) Computation of the per-pixel influence of each light and a la Retinex color correction (see Sec. IV-C).

In step 1 we gather an estimate of luminance through the computation of the luma image, then we take areas of the image that fall in the top 5% of it as *reference lights*.

In step 2 we exploit *chromatic information* to cluster the highlight regions according to their CIE - $Lab \Delta E_{00}$ measure. The clusters so formed are further checked for *spatial clustering* by means of Gap statistics, Euclidean distance and K-means clustering.

In step 3 each pixel in the image that is not part of a highlight area is corrected: every light in the image affects each pixel by an amount depending on its chromatic and spatial distance from the pixel itself.

In the following subsections we will describe in detail each phase of the method.

A. Highlight regions extraction

Given the input image $I(x, y)_{\text{ch} \in \{R, G, B\}}$ we determine the reference white $\mathbf{w} = [W_R \ W_G \ W_B]^T$ as:

$$w_{\text{ch}} = \max(I_{\text{ch}})^1 \quad (2)$$

An example input image can be seen in Fig. 2



Fig. 2. (a) Sample subject from the AR Face Database under neutral illuminant. (b) Same subject with yellow lateral illuminant added.

In order to obtain the estimates of the lights present in the scene we follow the method previously used in [19], but any other way of obtaining a good estimate for luminance may be used. We first gather the luma data using the formulation given by oRGB, expressed in (3):

$$L = 0.2990 \cdot R + 0.5870 \cdot G + 0.1140 \cdot B; \quad (3)$$

We then select the set of raster scan indices of those pixels in the top 5% of the normalized luma image as per (4).

$$\mathbf{T} = \left\{ \text{idx}(p) \mid p \in \frac{L}{\max(L)} \text{ and } p > 0.95 \right\} \quad (4)$$

Next, we create another RGB image M used to store the values of “top luma” pixels. Mathematically this is described in (5).

$$M(i) = \begin{cases} I(i), & \text{if } i \in \mathbf{T} \\ \mathbf{w}, & \text{otherwise} \end{cases} \quad (5)$$



Fig. 3. (a) Illuminance map M , given the input in Fig. 2-b. (the background is inverted for visibility purposes). (b) Binarized and cleaned luma map.

In order to distinguish the reference white from other lights, we compute the sum of M over the three chromatic channels,

and normalize it on the sum of the RGB values of the reference (Von Kries) white $w_{\text{sum}} = \sum_{\text{ch}} W_{\text{ch}}$. This is described in (6).

$$S = \frac{\sum_{\text{ch}} M}{w_{\text{sum}}} \quad (6)$$

Any value below 1 in S represents a pixel of a light with color different from the reference white. Such lights will be the main source of information for our correction process.

Finally we perform region labeling in the following way. We first “inverse binarize” S so that any pixel with value of 1 becomes 0, and any other pixel assumes value 1. In order to clean the resulting binary image, spur and isolated pixels are eliminated (spur pixels being pixels with only one corner connectivity). Then every pixel of an isolated region is labeled with the unique integer value assigned to that region, yielding a map similar to that shown in Fig. 5 (a). Mathematically we obtain a vector $\mathbf{r}$ containing the description of $R = \langle \mathbf{r} \rangle$ distinct regions (the acute bracket operator indicates the cardinality).

While running the experiments it has been noticed that, for particular images, the top 5% of the luma image does not provide enough information and the segmentation step produces only a single class of highlight areas, resulting in a pure Von Kries method. In order to solve this problem we suggest the repetition of the thresholding with a lower value each time, until more than one class is found. We are currently investigating in order to find an automatic solution to the problem.

B. Chromatic and spatial clustering

Since it is important to take into account chromatic and spatial relationships that might tie different highlight regions to each other, we perform a two-tier clustering devised to leverage on such relationships.

We first cluster the regions in $\mathbf{r}$ according to the inter-region ΔE_{00} . We obtain the symmetric difference matrix $\mathbf{D}$, which elements are defined as $\mathbf{D}_{m,n} = \{\Delta E_{00}(h_m, h_n) \mid m, n \in \{1, \dots, R\}\}$. Then we start generating clusters by constructing a matrix whose elements are 1 if $\mathbf{D}_{m,n} \leq 0.5$ and 0 otherwise (this will yield a maximum intra-cluster difference of 1). Computing the row-wise (or column-wise) sum on the binary matrix and picking the maximum value tells us which element in the matrix will be the new cluster center. We add the new cluster to the cluster list $\mathbf{r}_\Delta$, and repeat the process with the remaining elements until there are no more areas available for clustering.

The second step in the clustering process is based on the estimation of the number of spatially related clusters inside a chromatic related cluster. The estimate is computed through the use of the Gap statistics applied to K-means clustering. A full description of the Gap statistics can be found in [20], we will limit ourselves to a brief algorithmic explanation, given in Fig. 4. Note that, for clearness sake, we will explain in this paragraph how to compute the dispersion measure $DM_{(k,b)}^*$ mentioned in the pseudo-code. We first suppose to have clustered the data into clusters $C_1, \dots, C_k$, denoting with C_r the indices of observations in cluster r and with n_r the

Fig. 4. Spatial based clustering pseudo-code. Please refer to (7) for the computation of the dispersion measure.

```

for  $i = 1$  to  $R_\Delta$  do
  for  $k = 1$  to  $r_\Delta(i) >$  do
    Compute dispersion measure  $DM_k$ 
    for  $b = 1$  to  $B$  do
       $\text{Ref}(b, k) \leftarrow \text{random} \{x_{(b,k)}, y_{(b,k)}\}$ 
      Compute dispersion measure  $DM_{(k,b)}^*$ 
    end for
     $\text{Gap}(k) \leftarrow \frac{1}{B} \sum_{b=1}^B \log(DM_{(k,b)}^*) - \log(DM_k)$ 
     $l \leftarrow \frac{1}{B} \sum_{b=1}^B \log(DM_{(k,b)}^*)$ 
     $sd(k) \leftarrow \left[ \frac{1}{B} \sum_{b=1}^B \left( \log(DM_{(k,b)}^*) - l \right)^2 \right]^{1/2}$ 
     $s(k) \leftarrow sd(k) \cdot \sqrt{1 + \frac{1}{B}}$ 
    if  $k > 1$  then
      if  $\text{Gap}(k-1) \geq \text{Gap}(k) + s(k)$  then
         $\text{Guess}(i) \leftarrow k - 1$ 
      end if
    end if
  end for
end for

```

cardinality of C_r . If we compute the sum of intra-cluster pairwise distances as

$$IC_r = \sum_{i,i' \in C_r} d_{ii'}$$

then the dispersion measure DM_k is computed as

$$DM_k = \sum_{r=1}^k \frac{1}{2n_r} D_r \quad (7)$$

At the end of the whole clustering process we obtain something like what is shown in Fig. 5 (b).

Note that B is a parameter of the Gap method and can dramatically affect the computational performance. In our experiments we set $B = 25$, a number high enough to satisfy almost any need, but B should really be determined based on the confidence level desired for the application.

At the end of the spatial clustering we obtain a set of cluster indices, where each cluster represents an estimate of one light in the scene:

$$\mathbf{r}_{\text{gap}} = [h_1 \quad \cdots \quad h_{R_{\text{gap}}}] \quad (8)$$

C. Per-pixel color correction

In order to decide how to correct each pixel let us first state the assumptions that drove our findings:

- In [18] it was found that the structure used to scan the image (pixel “sprays”) would yield better results when its density was decreasing linearly with respect to the distance from the center.
- According to Hering’s theory no color looks red and green at the same time, nor can it look yellow and blue

(and so on, for all the possible color couple at opposite extremes of the hue wheel).

- We need the total “action” of the lights (intended as in Section IV) on a non-light pixel to be equal to that of a light areas on its own pixels.

These assumptions will be thoroughly explained in the rest of this section. Still, let us first define some variables for clarity: we will be considering a pixel to be corrected p_i , where i is the linear index of the pixel in the image ($i \in \{1, 2, \dots, \text{width}(I) \cdot \text{height}(I)\}$), and an highlight area h_j , where j is the linear index in the set of highlight clusters obtained after Fig. 4.

1) *Light influence with respect to chromatic and spatial distances:* We define a first parameter, $\alpha_{i,j}$, to be dependent on the Euclidean distance between the p_i and the centroid of h_j . Let Δ_{euclid} be the Euclidean distance, then

$$\alpha_{(i,j)} = 1 - \frac{\Delta_{\text{euclid}}(p_i, h_j)}{\text{diag}} \quad (9)$$

where diag is the diagonal of the image..

Similarly to α we define a second parameter, $\beta_{i,j}$, to be dependent on the distance between the hue value of p_i and the average hue value of h_j (note that that intra-cluster ΔE_{2000} differences are under the JND threshold). Let Δ_{hue} be the absolute hue difference ($\Delta_{\text{hue}} \in [0, 180]$), then:

$$\beta_{(i,j)} = 1 - \frac{\Delta_{\text{hue}}(p_i, h_j)}{180} \quad (10)$$

2) *Correction factor:* Given the definitions of α and β , we define the correction factor for a given pair of indices $\{i, j\}$ as:

$$\text{cf}_{(i,j)} = \alpha_{(i,j)} \cdot \beta_{(i,j)} \quad (11)$$

Such formulation has two desirable properties:

- $\text{cf}_{(i,j)} = 0$ if and only if either p_i and h_j are at opposite ends of the image along the diagonal, or if the hue difference is 180° (e.g. p_i is red and h_j is green)
- $\text{cf}_{(i,j)} = 1$ if and only if the distance between p_i and h_j is null and the hue difference is also null (true only for pixels which are part of the highlight area).

3) *Unitary light action:* Any pixel part of an highlight region will be corrected to its own local white (the region maximum on each of the color channels). From a mathematical point of view let $\text{cf}_{(i,1)}, \dots, \text{cf}_{(i,j)}$ be the correction factors for pixel p_i , and let p_i be part of the highlight area h_k . Then

$$\begin{cases} \text{cf}_{(i,j)} = 1, & \text{if } j = k \\ \text{cf}_{(i,j)} = 0, & \text{otherwise} \end{cases}$$

This leads us to compute the following sum:

$$\sum_{j=1}^{R_{\text{gap}}} \text{cf}_{(i,j)} = 1 \quad (12)$$

We want (12) to hold even for pixels that are not part of a highlight region: the total correcting action of the lights on a given pixel must be equal to 1.

4) *Per pixel correction*: In the end, given pixel p_i and regions r_{gap} , we compute the correction factor for each region as per (11). This gives us a vector of coefficients $\mathbf{cf} = [\mathbf{cf}_{i,1}, \dots, \mathbf{cf}_{i,R_{\text{gap}}}]$.

Since the pixel is given, we will omit the i index in the following equations for ease of understanding.

In order to comply to (12) we then normalize the vector in the following way:

$$\tilde{\mathbf{cf}}_{(j)} = \frac{\mathbf{cf}_{(j)}}{\sum_{j=1}^{R_{\text{gap}}} (\mathbf{cf}_{(j)})} \quad (13)$$

We also build the matrix of light colors (white points) as indicated next:

$$\mathbf{WP} = \begin{bmatrix} \max(h_{(1,R)}) & \max(h_{(1,G)}) & \max(h_{(1,B)}) \\ \vdots & \vdots & \vdots \\ \max(h_{(J,R)}) & \max(h_{(J,G)}) & \max(h_{(J,B)}) \end{bmatrix} \quad (14)$$

where $J = R_{\text{gap}}$ (for compactness reasons) and the maximum over a region h_j is defined as the maximum value of its member pixels.

The final per-pixel correction is performed according to the following equation:

$$\tilde{p}_{(ch)} = \frac{p_{(ch)}}{\sum_{j=1}^{R_{\text{gap}}} (\mathbf{WP}_{(j,ch)} \tilde{\mathbf{cf}}_{(j)})} \quad (15)$$

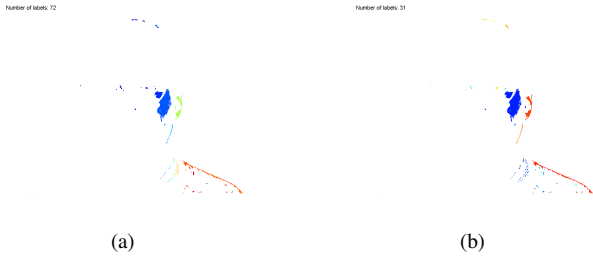


Fig. 5. Pseudo color maps of the labels before (a) and after clustering (b). Due to clustering the number of estimate lights has decreased from 72 to 31.

V. EXPERIMENTAL SETUP AND RESULTS

Since this work stemmed from research on skin tone color correction, even though it turned out into a general method, the experiments were run on images extracted from the “AR Face Database” [21]: this database has the interesting property of reproposing the same subjects under different viewing conditions (lighting, occluding elements, etc).

We chose two images per subject, one with neutral lighting and one with a distinct non-uniform color cast. We then applied our color correction method to the image with the color cast, and verified the chromatic difference with respect to the neutrally illuminated image. On a sidenote we have to mention that the database only offers left, right or “left and right” yellowish additional illumination and in such cases



Fig. 6. Corrected version of the input in Fig.2-b.

TABLE I
AVERAGE ORGB DIFFERENCES. THE

	Y	C_{yb}	C_{rg}
Non corrected	-26.68	-0.0808	0.009
$C_{yb}C_{rg}$ only	-29.56	0.0201	0.0202
Proposed method	-28.28	0.0178	0.0157

the exposure is far from ideal. The absence of multiple light colors and the excessive highlight clipping that occurs make consistent testing a lot harder.

Table I shows the average difference in the oRGB chroma channels between the ground truth (neutrally illuminated subjects), the subjects with added color cast, and the subjects with color cast and color correction applied. The results from the proposed method are compared to a simplified algorithm that performs the correction by using only a single light to correct each pixel, selecting the one with the minimum Euclidean distance from the pixel in $C_{yb}C_{rg}$.

From the numeric analysis it is apparent how the current choice of parameters tends to produce over corrected skin colors (the final skin tone is blue-ish), although the absolute average chromatic distance is reduced. The method performs marginally better than its simplified version: the computational cost is much higher, but we believe that in more complex situations the extra processing will lead to significantly better results.

Finally it is important to note that, being the correction inspired by Retinex, this method will always increase the lightness of the pixel under examination (although not dramatically since we limit the possible white reference to already luminous parts of the image) and that repeatability is not granted, due to the use of a statistical measure for clustering.

VI. CONCLUSIONS AND FUTURE WORK

As far as the authors’ knowledge is concerned this is the first time that an automatic method for color correction taking

into account both Land's Retinex and Hering's opponent color theories is devised.

The proposed method is a generic color correction algorithm theoretically able to deal with multiple light sources, and it is based on the assumption that any area of an image that has a high enough estimate of luminance should be considered as a light. Any light that differs from the image reference white, the global white point, influences all other pixels according to both Retinex and the opponent colors theories.

Tests have been conducted in the field of skin color correction, albeit with just one non-uniform light. The experiments show how the rendition of skin color improves notably after the proposed method is applied, but also indicate how the method needs further research and tuning to achieve better results.

Further research on the topic will regard the estimation of the luma threshold in the case the method reverts to Von Kries processing, the possibility of merging the two clustering steps, estimation of a proper value for the parameter B , and evaluation of the performance of different formulations of the correction factor.

Further testing will be carried out on more challenging databases, especially to test the method ability to correct multiple illuminants, and the results will be compared to other state of the art algorithms.

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